

JACOB TAM

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EDUCATION

Stephen J. R. Smith Faculty of Engineering and Applied Science, Queen's University
BEng – Electrical Engineering

Kingston, Ontario
Class of 2028

- **GPA:** 3.8/4.3 (Dean's Scholar All Years)
- **Varsity Tennis Team:** Placed 3rd at the OUA provincial championships in 2025-2026.
- **Relevant Coursework:** Microprocessor Interfacing & Embedded Systems, Numerical Methods & Optimization, Probability & Random Processes, Data Structures & Algorithms, Solid-State Devices, Digital Systems, Electronics II

PROFESSIONAL EXPERIENCE

Hydro One

Mississauga, Ontario

AMIO Network Equipment Specialist Intern

May 2026 – Present

- **Hardware adaptation:** Redesigned base and collector power circuitry during supply shortages (120 V AC to 12 V DC) with an integrated Microhard modem, keeping field deployment on schedule and saving over \$100 per unit.
- **Modem deployment:** Configured and shipped 450+ field modems in one month for Ontario's advanced metering network, adding SIMs and validating port forwarding, IPv4/IPv6 and SureLink over SSH/Citrix.
- **Vendor qualification:** Power-tested, flashed and signal-validated (RSSI -26 to -35 dBm) 200+ socket APs, access points, and collectors against the head-end system, hitting a ~95% pass rate and flagging defective units for RMA before deployment.

Forum Asset Management

Toronto, Ontario

Software Developer Intern

May – Aug 2025

- **Dynamic pricing:** Built a random forest model (scikit-learn, Pandas, NumPy) to recommend rent prices across Forum's portfolio of 35+ commercial and student-residence buildings (11,000+ units), trained on 2+ years of historical rent data with 30+ features (location, campus proximity, seasonality, and others).
- **Chatbot development:** Developed an agentic chatbot for an affiliated non-profit, routing users to partner resources based on demographics and needs, extending reach by 20%.
- **Marketing automation:** Designed an N8N marketing agent that analyzes competitor and news data to draft and publish LinkedIn content autonomously, lifting client engagement by 32%.

PROJECTS

Autonomous Robotic Taxi – Computer Vision & Embedded AI Project

Jan – Apr 2026

- **Computer vision:** As perception lead, trained a TensorFlow Lite traffic-sign detection model on 1,000+ images per class, quantized onto a Coral Edge TPU, improving accuracy from 93% to 97%+.
- **System integration:** Wrote the code that brought line-following, heading-based A* pathfinding, fare-claiming, and turning logic together into a 20 Hz async control loop; as the only EE on the team, also wired the turn-signal and brake lighting system.
- **Competition:** Placed 3rd of 55 teams in total fare revenue (\$1,032.66).

Anti-Theft Package Smartbox – IoT Hardware/Software Project

Sept – Dec 2024

- **Smart hardware:** Engineered an IoT package box (Arduino, motion sensors, solenoid lock) with an SQL-backed web dashboard and a live video feed for remote monitoring and secure access.
- **AI security:** Built a facial-recognition module (OpenCV, DeepFace) at 98% intruder-detection accuracy with real-time theft alerts.

EXTRACURRICULAR EXPERIENCE

Queen's Hyperloop Design Team (QHDT)

Kingston, Ontario

Suspension Design Engineer

Sept 2024 – Jan 2025

- **Award-winning design:** Redesigned the Hyperloop pod suspension system, recognized with the most awards nationally at Hyperloop Global 2024.
- **Efficiency improvements:** Optimized the suspension and clamping mechanism via iterative CAD/FEA analysis, reducing drag by 56% (validated through testing).

SKILLS

Programming: Python, C, Java, JavaScript/TypeScript, SQL, MATLAB, VHDL

Machine Learning & Data: scikit-learn, TensorFlow, OpenCV, Pandas, NumPy, Matplotlib, Power BI

Hardware: Raspberry Pi, Arduino/microcontrollers, Coral Edge TPU, sensor integration, RF/RSSI testing, PCB prototyping, soldering, lab instrumentation, AC-DC power circuits

Tools & Platforms: Git, Linux, SSH, REST APIs, React, Django, Citrix, SolidWorks, LTSpice, PowerApps, N8N

Languages: English, French (fluent); Ontario Certificate in Bilingual Studies